HOW THE NUMBERS WERE MADE

How we measured COSMOS

How the numbers on cosmos.devliam.com were produced, scored, and where they stop being reliable.

×23
fewer tokens sent per question

×11
faster answers

97%
of searches put the right memory in the top 3

8 open models · 298 questions each · about 200 memories · 3 random draws · 2 x Tesla T4

00 · THE SHORT VERSION

What we did, in one minute

COSMOS gives an AI assistant a memory that lives outside its head. The question we wanted to answer is simple: **does an assistant do better when it reads only the few notes COSMOS picks, instead of every note at once?**

1 **We wrote a test**
Notes and questions, some about made-up people so no model can guess the answer.

2 **We asked 8 AI models**
The same questions, three ways: no notes, all notes pasted, or only the 3 notes COSMOS finds.

3 **We counted**
Right answers, how much the model had to read, and how long it took.

×23	×11	97%	4 · 2 · 2
fewer tokens read per question	faster answers	right memory in the search top 3	models won · tied · lost vs the full file

THE HONEST PART

COSMOS is **not better everywhere**. Against pasting the whole file it wins on 4 models, ties on 2 and loses on 2 (Llama 3.2 3B and Qwen3.5 4B). This document lists exactly how the test works, so you can judge the numbers yourself.

How to read this document

- Boxes titled **In plain words** give the idea without the details.
- Figures are drawn from the real result files, not from the summary text.
- The last sections list what the test does **not** prove.

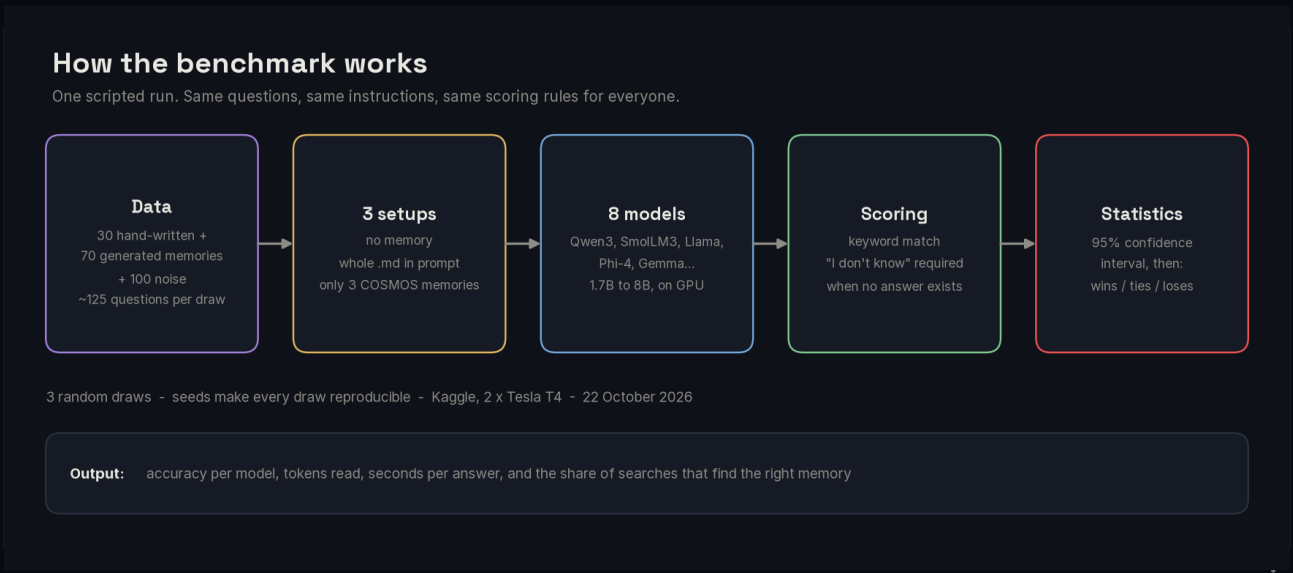
01 · THREE MEASUREMENTS

What was measured

IN PLAIN WORDS

We checked three things: does the **search** find the right note, does an **AI answer correctly** with it, and does our own **search model, Polaris**, do better than the ones it replaces.

PART	NAME	WHAT IT MEASURES	USES AN AI MODEL TO ANSWER?
A	Search alone	Does the search find the right memory? How fast, as memory grows from 150 to 1,700 entries.	No
B	Answers with an AI	Does a model give the right answer with no memory, the full file, or COSMOS?	Yes, 8 models
C	Polaris vs other search models	Does the fine-tuned search model find memories better than the models it replaces?	No



The path every number takes, from test data to verdict.

02 · TEST DATA

What the models were asked

IN PLAIN WORDS

A mix of real-looking notes and generated ones about **made-up people** (invented surnames, companies, pets). A model cannot answer from what it already knows: it can only answer from the memory it is given.

Everything is in French, because COSMOS is mainly used in French. The data has three sources.

Hand-written memories and questions

- **30 memories** written by hand (not generated), in six categories: facts (10), preferences (5), project context (5), decisions (4), tasks (3), open questions (3). File: `benchmark/dataset.json`.
- **38 questions** written by hand: 23 direct (same words as the memory), 5 rephrased (same meaning, other words), 2 about an update (the newest value is expected), 2 about history (the old value is expected), 6 with no answer in memory.

Generated cases

- For each random draw (a "seed"), `benchmark/synth.py` generates **60 cases** about a made-up person, with random company names, pet names, codes and dates.

- Six case types rotate: pet name, job, city, birthday, safe code, and move (old city, then new city). A move creates two memories, the second one marked as replacing the first. Half of them also get a history question.
- Every third case adds a question with no answer (phone number, car color, favorite dish, height, salary).
- The seed fixes every random choice, so any draw can be regenerated identically.

Noise

- Unrelated memories built from templates ("X lives in Y for N years", "appointment with X on a day at an hour"...), inserted at random places without changing the order of the real memories.
- For the AI tests: 100 noise memories. Total: 30 + 70 generated + 100 noise = **about 200 memories**. The search-only series goes from 50 to 1,600 noise memories (150 to 1,700 in total).

	PER DRAW	3 DRAWS (AI TESTS)
Memories (with 100 noise)	about 200	same data, 3 different random draws
Questions	about 125 (about 99 with an answer, about 26 without)	375 (298 with an answer, about 77 without)

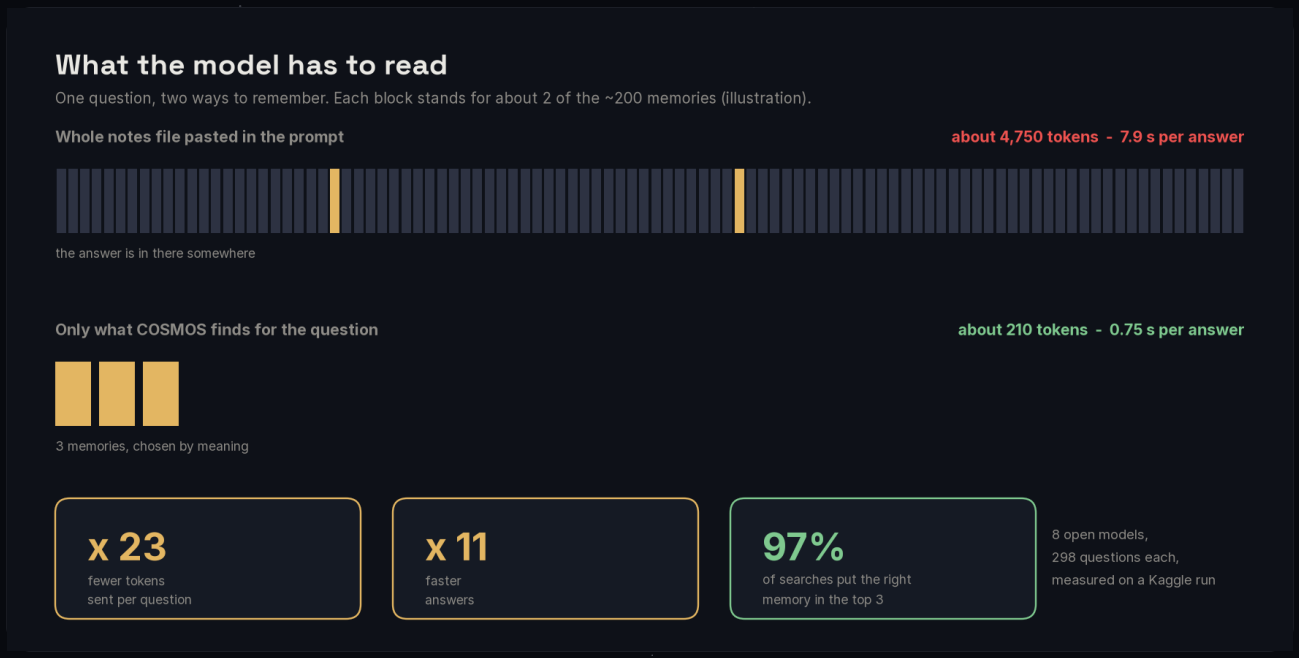
03 · THREE SETUPS

What the model gets to read

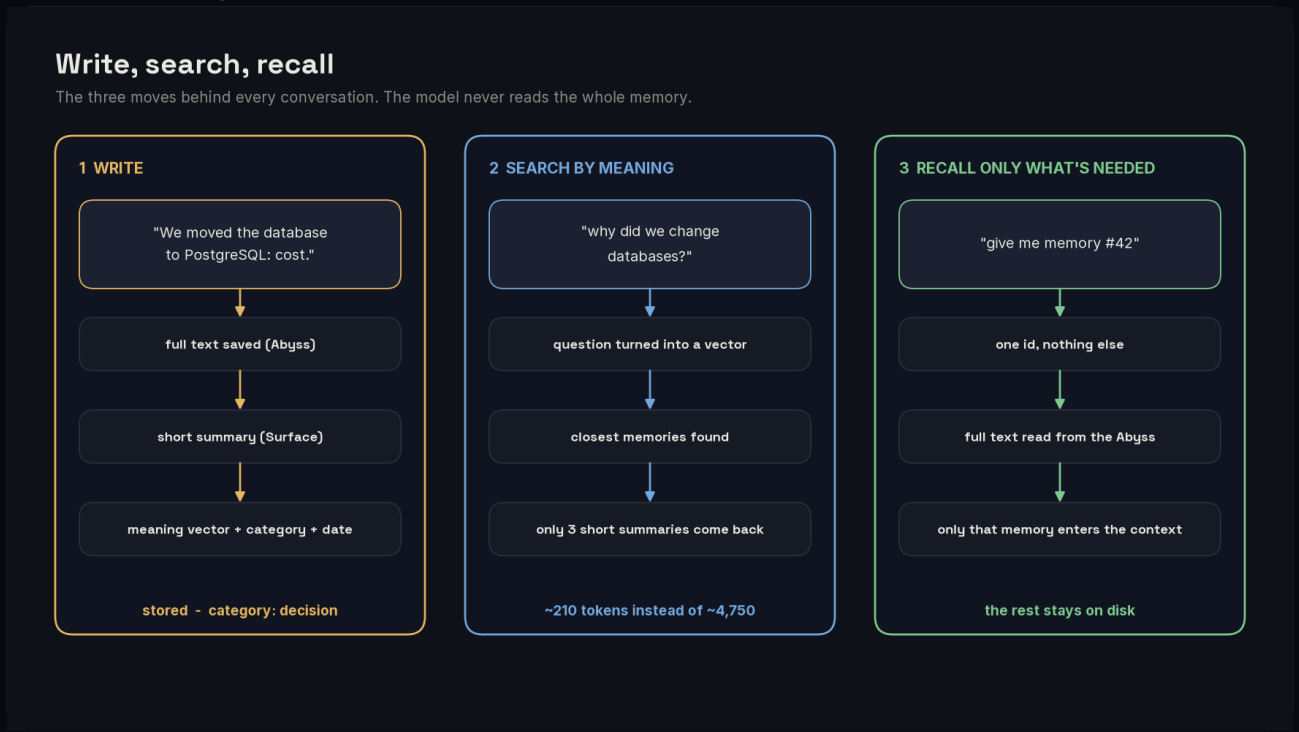
IN PLAIN WORDS

Same question, same instructions, same model. The only thing that changes is the memory it is given: **none**, **everything**, or **only what COSMOS finds**.

SETUP	WHAT THE MODEL READS
none	"(no memory)". The floor: it shows the model cannot answer by itself.
full .md	All memories pasted in the prompt as a list, numbered in the order they were written.
cosmos	Only the 3 memories COSMOS finds for the question, in chronological order, as full text. If COSMOS knows a memory was replaced by a newer one, the line says so.



The two memory setups side by side. Each block stands for about 2 memories (illustration).



Background: how COSMOS finds memories. The "cosmos" setup uses step 2 to pick the 3 memories.

Each memory line looks like `- [memory no. 12] text` or `- [memory no. 12, replaced by a newer one] text`. A higher number means a more recent memory.

A BIAS YOU SHOULD KNOW ABOUT

Only COSMOS can mark a memory as **replaced**, because it detects versions of the same fact. The full file has the numbers (a model can reason that higher means newer) but no such mark. This helps COSMOS on update questions. It is part of what COSMOS does, but it is not a like-for-like comparison on that category.

Instructions given to the model (system prompt)

```
Tu es un assistant qui connait l'utilisateur grace a une memoire. Reponds en francais, en une phrase courte, uniquement a partir de la memoire fournie. Si la reponse n'est pas dans la memoire, reponds exactement : Je ne sais pas. Chaque souvenir a un numero : plus il est grand, plus il est recent. Si deux souvenirs se contredisent, le plus recent fait foi, sauf si la question porte sur le passe.
```

English: "You are an assistant who knows the user thanks to a memory. Answer in French, in one short sentence, only from the memory provided. If the answer is not in the memory, answer exactly: I don't know. Each memory has a number: the bigger, the more recent. If two memories contradict each other, the most recent one wins, unless the question is about the past."

04 · MODELS

Who answered, and how

MODEL	MAKER	SIZE	HUGGING FACE ID
Qwen3 1.7B	Qwen	1.7B	Qwen/Qwen3-1.7B
SmolLM3 3B	Hugging Face	3.1B	HuggingFaceTB/SmolLM3-3B
Llama 3.2 3B Instruct	Meta	3.2B	unsloth/Llama-3.2-3B-Instruct
Phi-4 mini Instruct	Microsoft	3.8B	microsoft/Phi-4-mini-instruct
Qwen3 4B Instruct 2507	Qwen	4B	Qwen/Qwen3-4B-Instruct-2507
Gemma 3 4B it	Google	4.3B	unsloth/gemma-3-4b-it
Qwen3.5 4B	Qwen	4.7B	Qwen/Qwen3.5-4B
Qwen3 8B	Qwen	8.2B	Qwen/Qwen3-8B

- **Same settings for all:** the model's own chat template, deterministic generation (no random sampling, so the same prompt always gives the same answer), at most 96 new tokens, "thinking" mode turned off where the model has one. A leftover thinking block is removed before scoring.
- **Precision:** 16-bit floats first. If the model then produces gibberish on a test question, the run retries in bfloat16, and stops with an error if neither works.
- **Hardware:** Kaggle with 2 Tesla T4 GPUs, one GPU per model by default. Python 3.13, PyTorch 2.11, Transformers 5.20, fastembed 0.9 (recorded in `benchmark-cosmos/meta.json` for the second pass).
- **Two passes:** Qwen3 4B and Qwen3 8B ran in a second pass with the same settings. Their full-file prompt (about 5,000 tokens) did not fit in GPU memory in one go, so long prompts were read in pieces of 1,024 tokens: the same computation, with less memory.

05 · SCORING

How an answer counts as right

IN PLAIN WORDS

An answer is right if it contains the expected value. If the memory has no answer, the **only** right reply is "I don't know": inventing something counts as wrong.

A worked example (illustration, not taken from the test set)

MEMORY	QUESTION	MODEL ANSWER	VERDICT	WHY
"Jade Zovi moved: she now lives in Nantes."	Where does Jade Zovi live now?	"Jade vit maintenant a Nantes."	Right	Contains the expected word, "nantes".
same	same	"Je ne sais pas."	Wrong	The answer existed; the model gave up.
(nothing about her salary)	What is Jade Zovi's salary?	"Je ne sais pas."	Right	No answer exists: saying so is correct.
same	same	"About 3,000 euros."	Wrong	Made up.

- **With an answer:** correct if the answer contains every expected keyword group (accents, case and punctuation ignored) and does not say "I don't know". Numbers match as whole numbers: "4" does not match inside "14", and "4 500" equals "4500".
- **With no answer:** correct only if the model says it does not know (a list of phrases such as "I don't know", "no information", "not mentioned").
- **Headline accuracy** counts only questions with an answer (298 per model and setup). The no-answer questions are reported separately as "abstention".
- **Keyword scoring is strict and consistent, but not human.** It can mark a good answer wrong if it uses other words, and a bad answer right if it happens to include the keyword.

06 · STATISTICS

The numbers, in plain words

MEASURE	IN PLAIN WORDS
Accuracy	Right answers divided by questions asked, with a 95% Wilson confidence interval (the brackets in the table, the thin line on the bars).
Confidence interval	The range where the true score probably lies. If we ran the test again, the new score would most likely land inside it.
Wins / ties / loses	COSMOS vs the full file on the same model. A gap counts only if it is larger than the combined uncertainty (square root of the two interval half-widths squared and added). Otherwise: tie.
Tokens	The size of the whole prompt, counted with each model's own tokenizer, averaged over all questions.
Time per answer	Seconds to produce the answer on a Tesla T4, including reading the prompt.
Recall@k	Share of questions where the right memory is among the first k search results (k = 1, 3, 5).
MRR	Mean of 1 divided by the rank of the right memory. 1 means always first.
Abstention (search only)	Best accuracy at saying "nothing found" when the best match scores below a threshold. The threshold is chosen on the test itself.

Search-only numbers are averaged over the random draws (5 draws for part A); the spread is the standard deviation between draws. Search time is the median of one search, measured on the Kaggle CPU. Automatic category is the share of the 30 hand-written memories COSMOS files under the category written in the dataset.

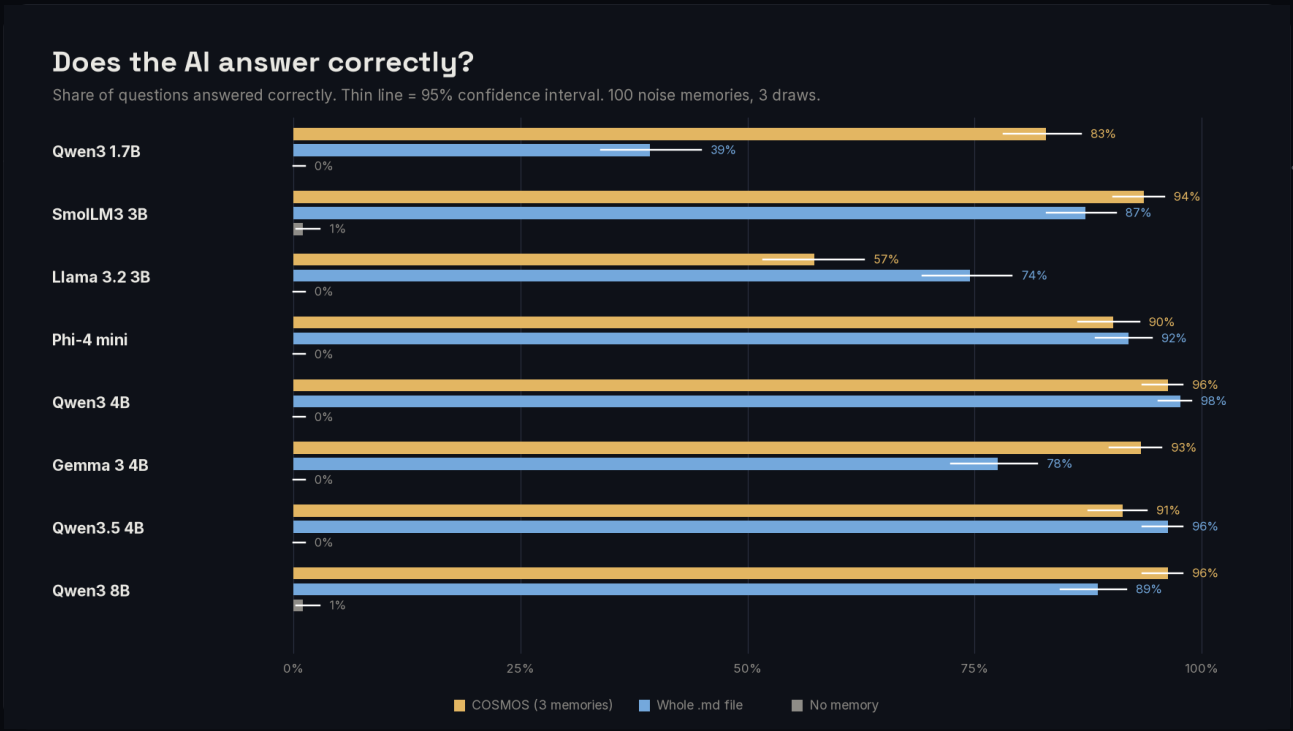
07 · RESULTS

What came out

HOW TO READ THE CHART

Gold is COSMOS, **blue** is the whole file pasted in the prompt, grey is no memory. A longer bar is better. The thin white line is the uncertainty: when two lines overlap, the gap is not trustworthy.

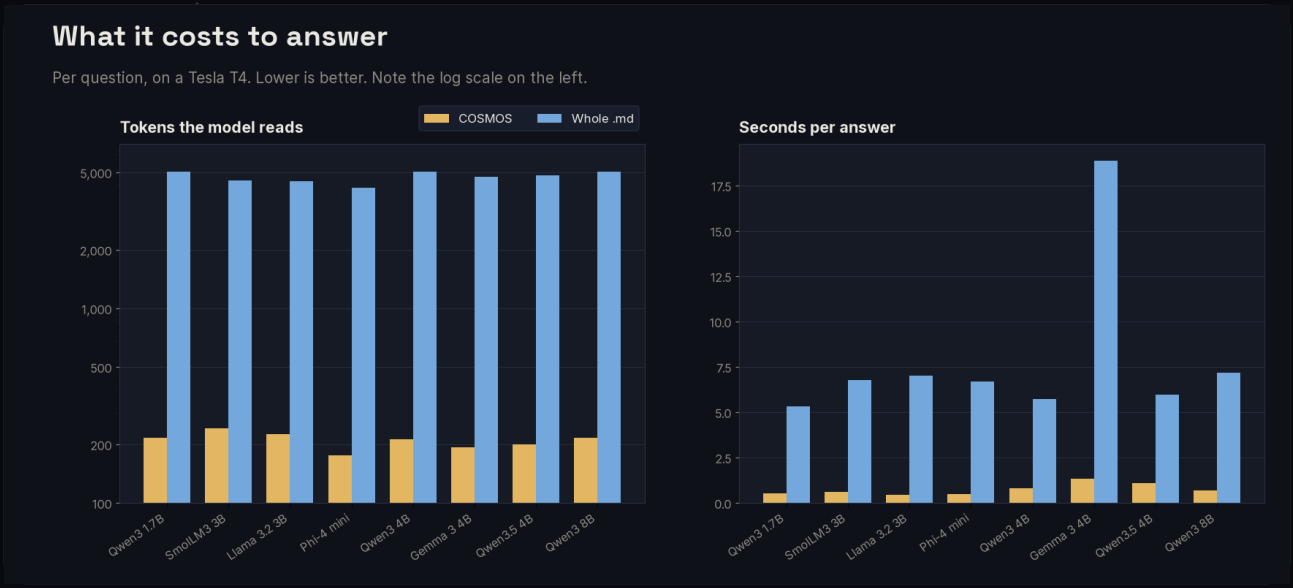
Part B: answers with an AI (100 noise memories, 3 draws)



Accuracy per model with 95% confidence intervals.

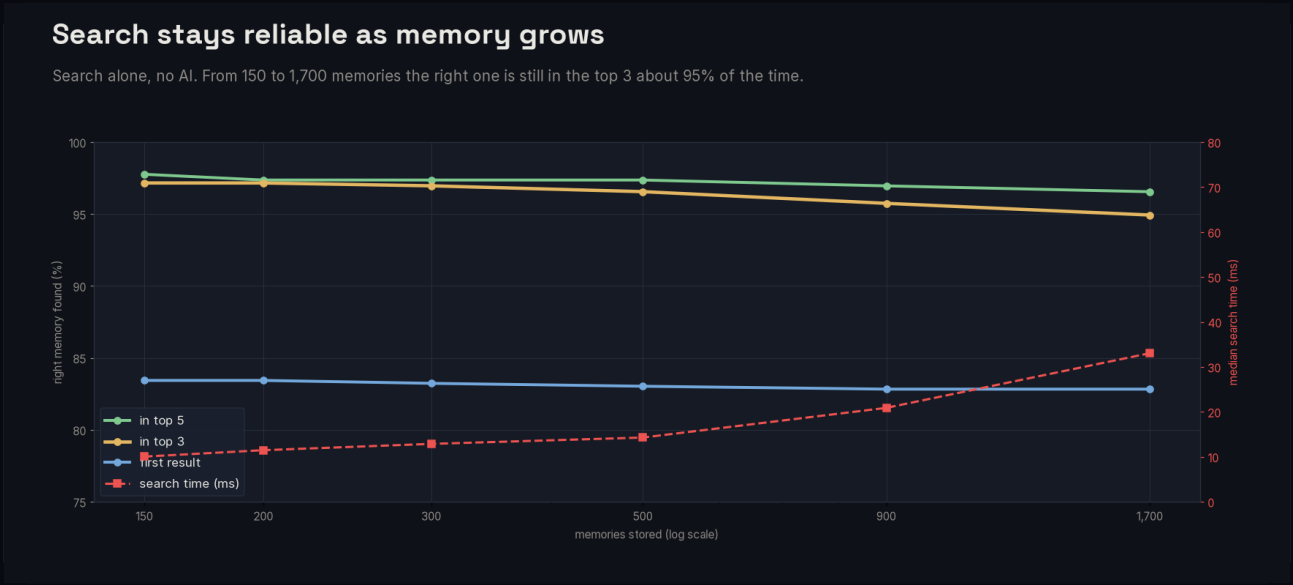
MODEL	COSMOS	FULL .MD	NO MEMORY	VERDICT	TOKENS (C / FULL)	SECONDS (C / FULL)
Qwen3 1.7B	83% [78-87]	39% [34-45]	0%	COSMOS	216 / 5,060	0.5 / 5.3
SmolLM3 3B	94% [90-96]	87% [83-91]	1%	COSMOS	243 / 4,530	0.6 / 6.8
Llama 3.2 3B	57% [52-63]	74% [69-79]	0%	Full .md	227 / 4,514	0.5 / 7.0
Phi-4 mini	90% [86-93]	92% [88-95]	0%	Tie	176 / 4,171	0.5 / 6.7
Qwen3 4B	96% [94-98]	98% [95-99]	0%	Tie	212 / 5,056	0.8 / 5.7
Gemma 3 4B	93% [90-96]	78% [72-82]	0%	COSMOS	193 / 4,752	1.3 / 18.8
Qwen3.5 4B	91% [88-94]	96% [94-98]	0%	Full .md	200 / 4,815	1.1 / 6.0
Qwen3 8B	96% [94-98]	89% [84-92]	1%	COSMOS	216 / 5,060	0.7 / 7.2

C = COSMOS. COSMOS searched with MiniLM during this run (Polaris came later). On average: about 210 tokens with COSMOS against about 4,750 with the full file (23 times fewer), and about 0.75 s against 7.9 s per answer (11 times faster). Source: `benchmark-cosmos/SUMMARY.md`.



What it costs to answer: tokens read and seconds per answer, per model.

Part A: search alone (memory growing from 150 to 1,700 entries, 5 draws)



Search quality and speed as the memory grows.

MEMORIES	TOP 1	TOP 3	TOP 5	MRR	SEARCH TIME
150	83.4%	97.2%	97.8%	0.903	10 ms
300	83.2%	97.0%	97.4%	0.899	13 ms
500	83.0%	96.6%	97.4%	0.897	14 ms
900	82.8%	95.7%	97.0%	0.894	21 ms
1,700	82.8%	94.9%	96.6%	0.891	33 ms

Search alone used the earlier search model (MiniLM). Other measured values: the newest memory ranked above the old one 70% of the time; "no answer" detected 85% of the time; automatic category right 21 to 39%.

08 · POLARIS

The search model

IN PLAIN WORDS

Polaris is the part of COSMOS that decides which notes are "close" in meaning to your question. We trained it on memories written for COSMOS, then checked it against the models it replaces.

Polaris is `intfloat/multilingual-e5-small` (118 million parameters, MIT license) fine-tuned on memories written for COSMOS.

Training

- **Data:** 7,040 memories written for this purpose, in 12 languages (English, Spanish, French, Italian, German, Portuguese, Dutch, Arabic, Chinese, Japanese, Turkish, Polish). Each has three different ways to ask for it, a near-miss memory that does not answer (same person, another fact), and for about half of them a newer version of the fact. Written with AI, not collected from real users.
- **Method:** full fine-tuning with a contrastive loss (MultipleNegativesRankingLoss): each question must land closer to its memory than to the near-miss and to the other memories of the batch. 26,029 training examples, 2 epochs, batch of 64, 2 x Tesla T4, about 10 minutes. 6% of the themes were kept out to check the model on memories it never saw.
- **Fair test:** the names and values used by the COSMOS benchmark were banned from the training data.
- **A first attempt failed:** it taught the model that the old version of a fact is a wrong answer. It then ranked updates first every time, but stopped finding old values and rephrased questions (right memory first in 45% of cases). The final Polaris treats both versions as right answers, and COSMOS puts the newest first using the date.

Comparison

Search alone, no AI, about 200 stored memories, 3 random draws (about 125 questions each).

SEARCH MEASURE	MINILM	E5-SMALL	POLARIS
Right memory ranked 1st	85.6%	87.3%	85.3%
Right memory in the top 3	97.6%	96.3%	**98.3%**
Ranking quality (MRR)	0.914	0.920	0.915
"Not in memory" recognized	84.6%	84.8%	**92.0%**
Newest version of a fact ranked 1st	91.7%	91.7%	86.1%

Polaris is not ahead everywhere. It ranks the right memory 1st slightly less often than the untuned model (2 or 3 questions out of 125, within chance), and it ranks the newest version of a fact 1st less often, because it treats old and new versions as equally right. COSMOS fixes that order afterwards using the date.

09 · LIMITS

What these numbers do not say

READ THIS PART

A benchmark is only as good as its weak points. These are ours.

LIMIT	WHAT IT MEANS FOR YOU
One memory size with AI	About 200 memories, 3 draws. Larger runs stopped when the GPU ran out of memory. The AI results do not show what happens at 1,700 memories.
Small and mid-size models only	1.7 to 8 billion parameters. Very large models read long prompts better, so the gap could be smaller.
French data, one prompt	Another prompt wording could change who wins.
Mostly generated data	70 of the 100 non-noise memories, all the noise and about 85 of the 123 questions per draw are generated. Real notes are messier.
Keyword scoring	No human or judge model. A good answer in other words can be marked wrong.
Settings tuned on the test	The "no answer" threshold and the version setting were chosen on this benchmark, so every model looks slightly better than on new data.
Update questions favor COSMOS	Only COSMOS can mark a replaced memory (section 03).
Small samples	About 125 questions per draw. Gaps of 1 to 2 points are within chance. The verdict uses confidence intervals, not a full test across models.
Synthetic Polaris training data	Written by an AI. Gains on real notes will likely be smaller.
Not a win everywhere	Against the full file COSMOS loses on Llama 3.2 3B (57% vs 74%) and Qwen3.5 4B (91% vs 96%).
Weak spots measured	Automatic categorization is right only 21 to 39% of the time; detecting "no answer" without an AI tops out around 85%.
Run by the author	No independent party has repeated it yet.

10 · REPRODUCE IT

Run it yourself

Everything is in the project repository (`github.com/DevLiamDb/COSMOS`). The seeds make every draw identical.

ITEM	WHERE
Test data (hand-written)	`benchmark/dataset.json`
Case generator	`benchmark/synth.py`
Runner: search, prompts, scoring	`benchmark/run.py`
Run every model, then report	`benchmark/run_all.py`, `benchmark/report.py`
Notebook used on Kaggle	`benchmark/kaggle_notebook.ipynb`
Raw results per model	`benchmark-cosmos/raw/*.json`
Every answer of every model	`benchmark-cosmos/tables/toutes_les_reponses.csv`
Machine, versions, settings	`benchmark-cosmos/meta.json`
Charts and tables	`benchmark-cosmos/report.html`

```
# search only, no AI
python -m benchmark.run --scales 50,100,200,400,800,1600 --seeds 0,1,2,3,4
# with one model
python -m benchmark.run --llm Qwen/Qwen3-8B --scales 100 --seeds 0,1,2
```

Numbers will differ slightly on other hardware (speed above all). Accuracy should match closely because generation is deterministic, but small differences between library versions are possible.

Set in Space Grotesk and Inter (SIL Open Font License). Colors and glyph from the COSMOS design system. Written with AI under the direction of Liam DB; see CREDITS.md in the repository.